Spatial Transcriptomics on Breast Cancer Dataset: Histology to Gene expression

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02 Literature Review

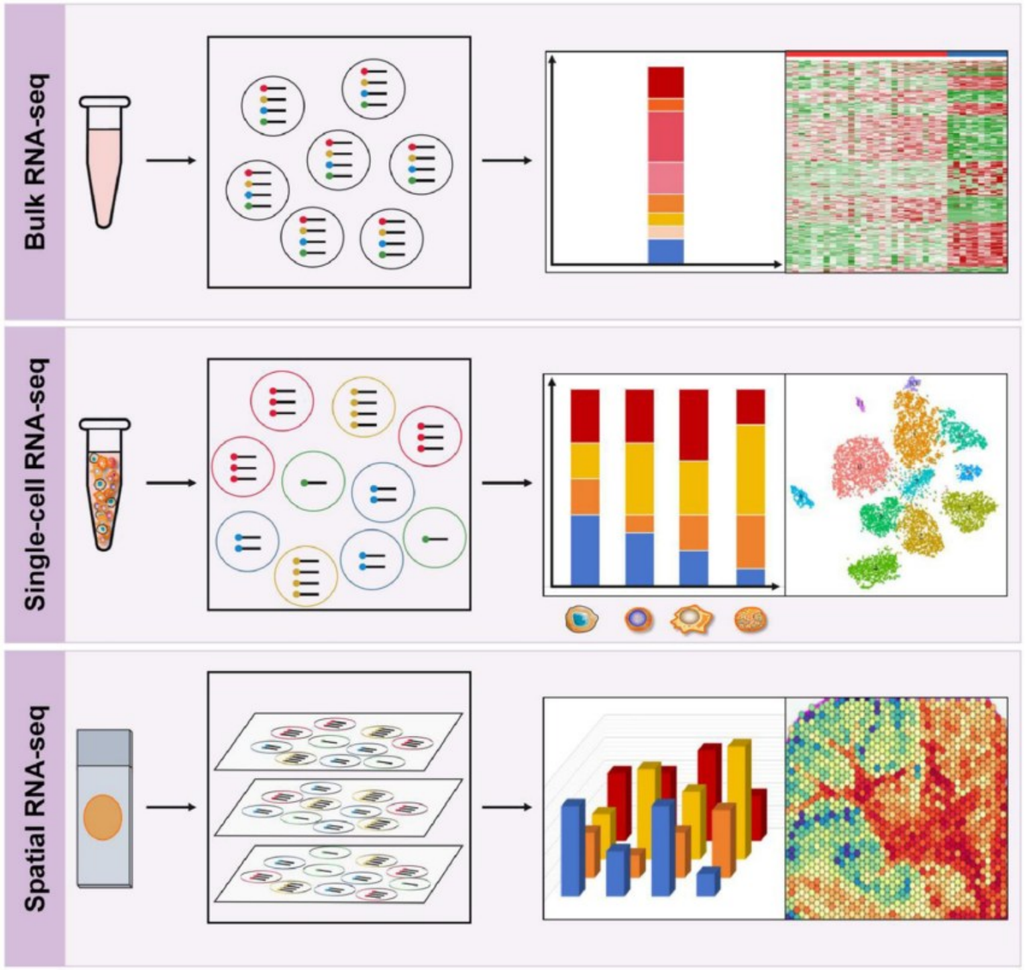
03 Dataset

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Transcriptomics Research Flow



Bulk RNA sequencing

Measures average gene expression across all cells, losing cellular heterogeneity.

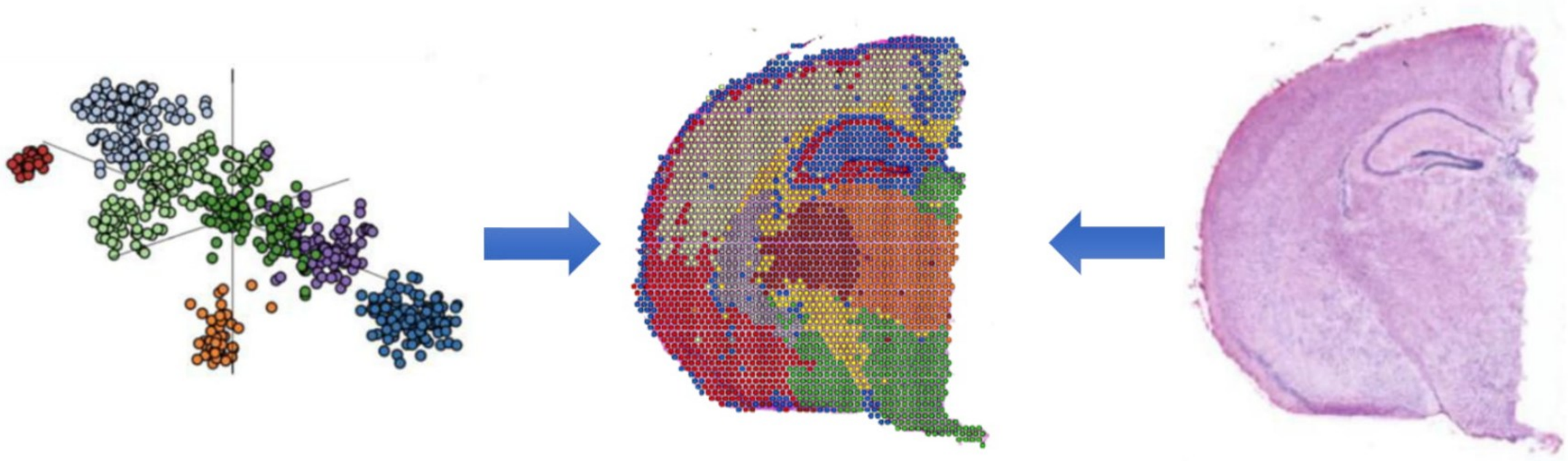
Single cell RNA sequencing

Profiles gene expression at single-cell resolution but lacks spatial information.

Spatial RNA sequencing

Preserves spatial context while measuring gene expression across tissue sections.

Spatial Transcriptomics



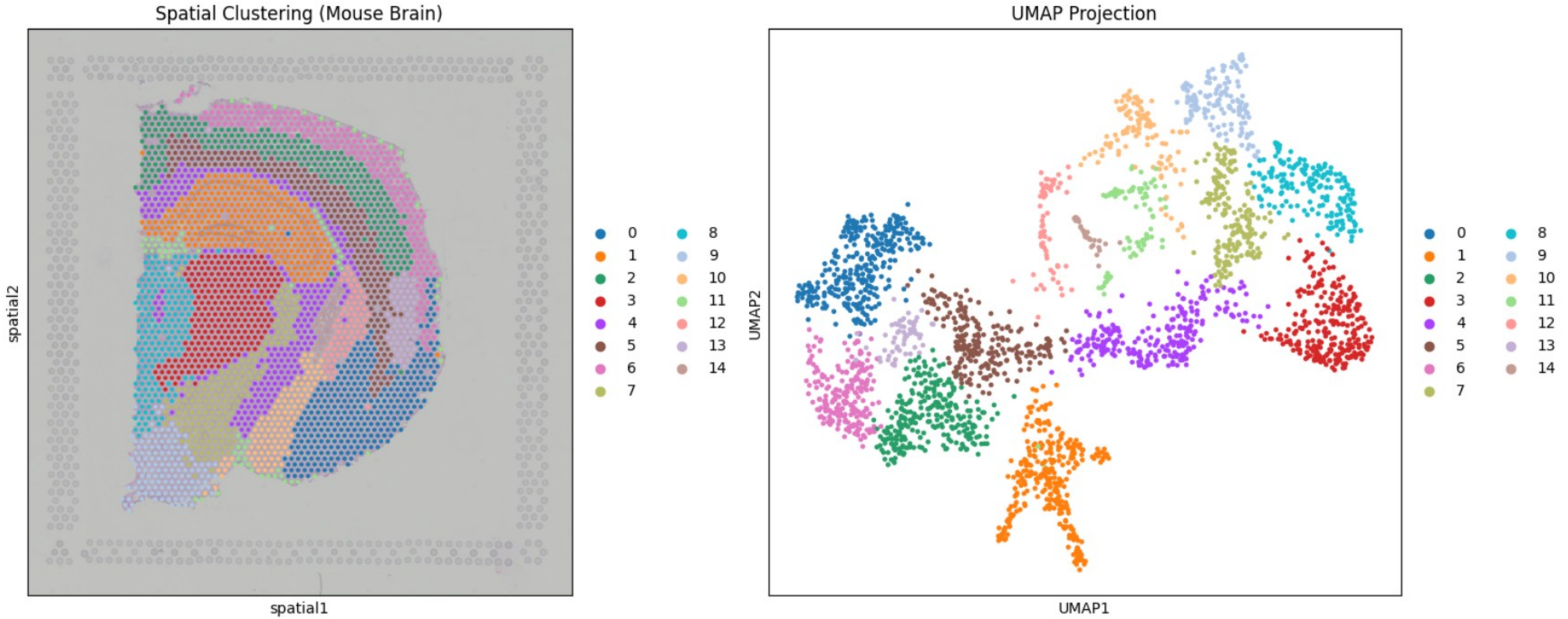
Single Cell Gene Expression

Spatially Resolved Gene Expression

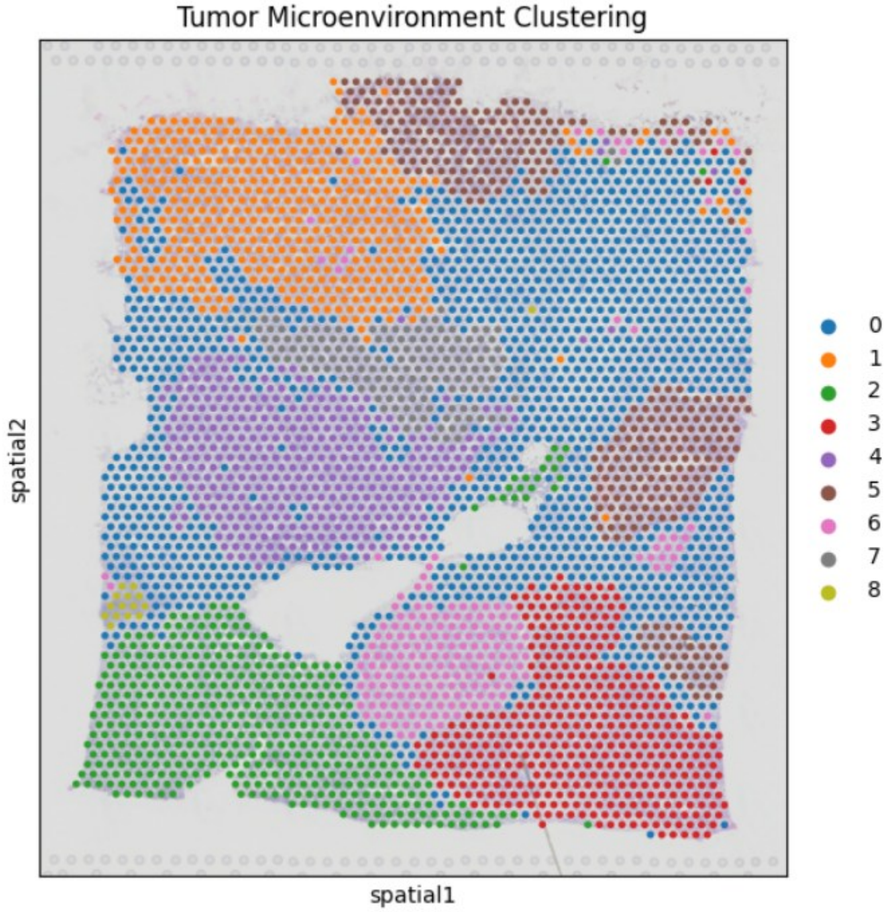
Tissue Section

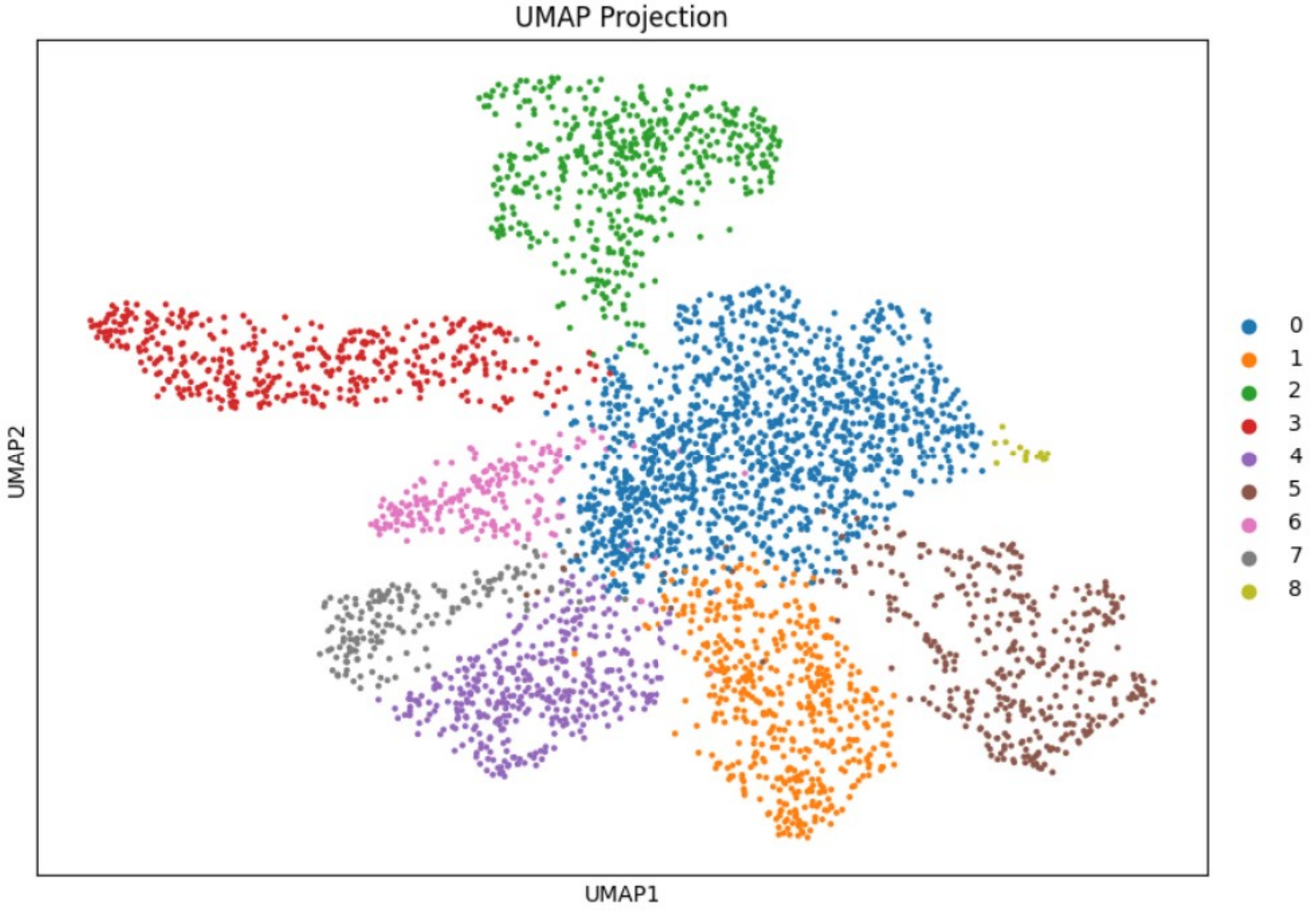
Adapted from 10x Genomics

Base Study - Mouse Brain Visium



Base Study - Tumor MicroEnvironment (TME)





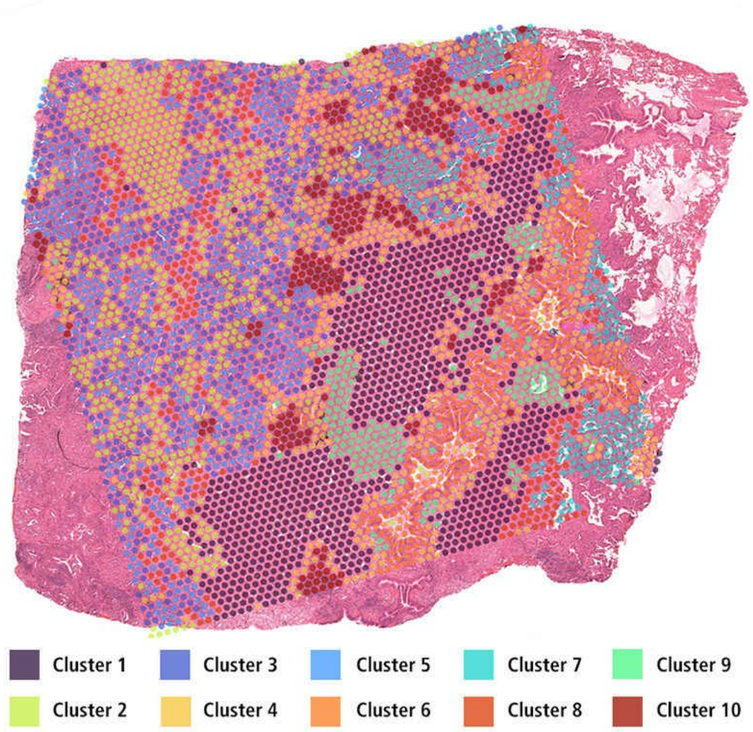
Research Background

Limitations of Current Spatial Transcriptomics

- Technologies such as 10x Visium are expensive
- Limited spatial resolution (~55 µm spots)
- Each spot contains multiple cells (~5–10 cells)
- True single-cell resolution remains challenging

Limitation of Histology

- H&E staining is inexpensive and routinely available
- Provides rich morphological information
- Lacks molecular and gene expression information



Literature Review: HisToGene

nature methods

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Spatial gene expression at single-cell resolution from histology using deep learning with GHIST

Xiaohang Fu, Yue Cao, Beilei Bian, Chuhan Wang, Dinny Graham, Nirmala Pathmanathan, Ellis Patrick, Jinman Kim & Jean Yee Hwa Yang ✉

Nature Methods **22**, 1900–1910 (2025) | Cite this article

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Abstract

The increased use of spatially resolved transcriptomics provides new biological insights into disease mechanisms. However, the high cost and complexity of these methods are barriers to broader application. Consequently, methods have been created to predict spot-based gene

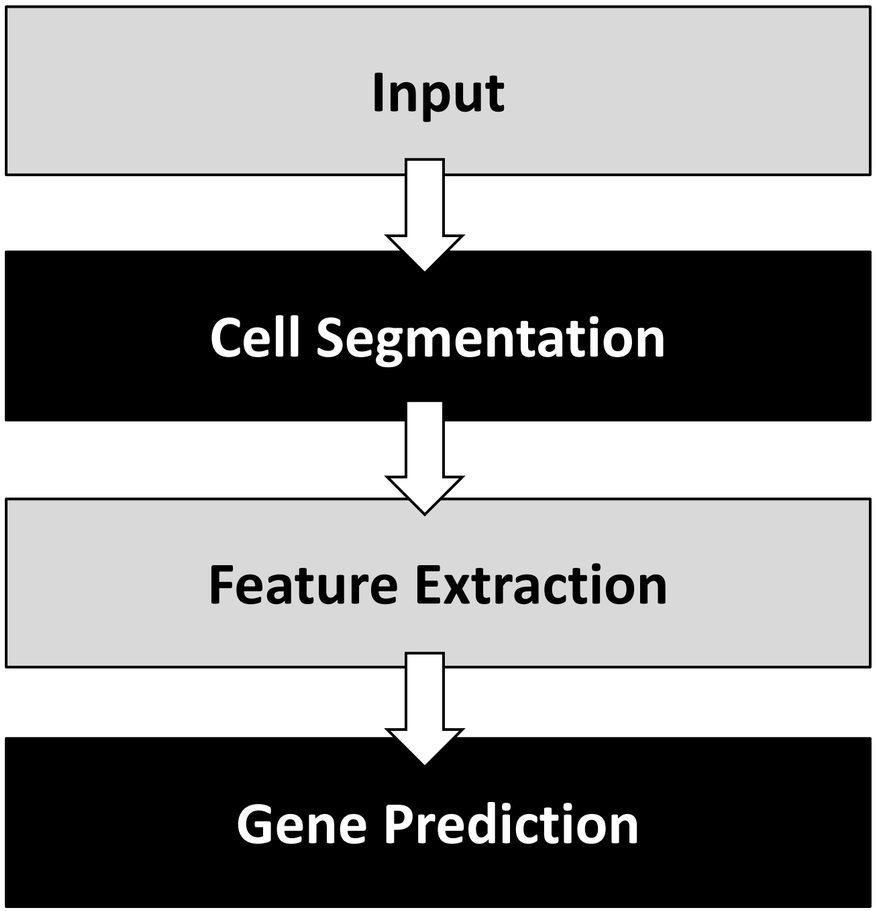
Research Objective

Predict single-cell spatial gene expression directly from routine H&E-stained histology images

Key Contributions

- Learn molecular patterns from histological features
- Enables large-scale tissue profiling from standard pathology slides
- bridges the gap between inexpensive histology images and high-resolution spatial transcriptomics

Literature Review: HisToGene



Whole-slide H&E image (WSI)

Pretrained cell segmentation model identifies individual cells

Vision Transformer (ViT)-based encoder extracts cellular features

Regression model predicts gene expression for each cell

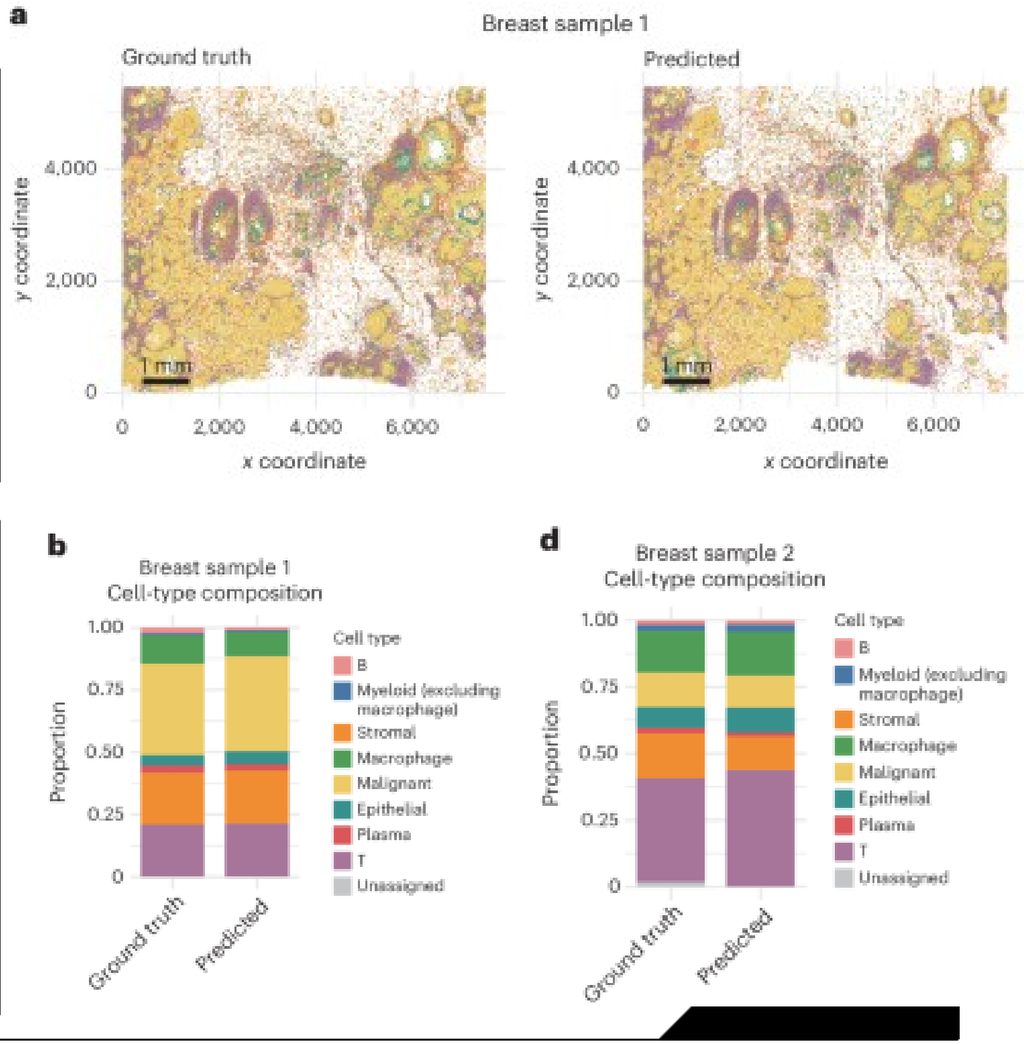
Literature Review: HisToGene

Performance Evaluation

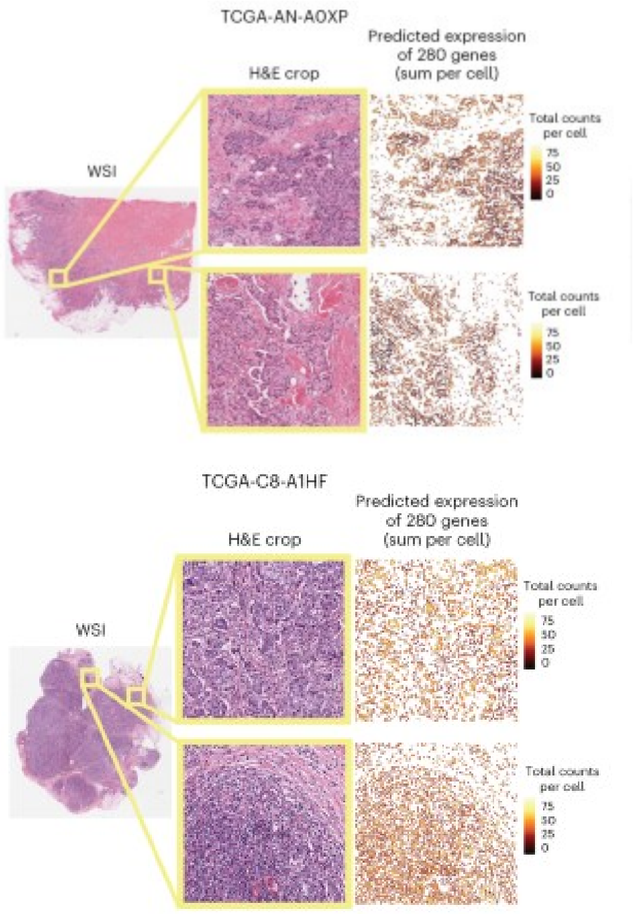
- Validated across multiple tissue types
- 10x higher spatial resolution than spot-based approaches
- Successfully predicted cell-type specific marker genes

Biological Insights

- Captured spatial heterogeneity within the tumor microenvironment
- Visualized immune-cell infiltration patterns
- Identified rare cellular populations



Literature Review: HisToGene



Significance

- Cost Efficiency
- High Resolution
- Scalability

Limitations

- Data Bounds (e.g. Visium spots)
- Low Expression Markers
- Generalization on unseen tissue

Future Direction

- High Resolution Platforms
- Multi-modal multi-omics
- Cross-tissue robustness

From Literature Review to Our Project

The Challenge

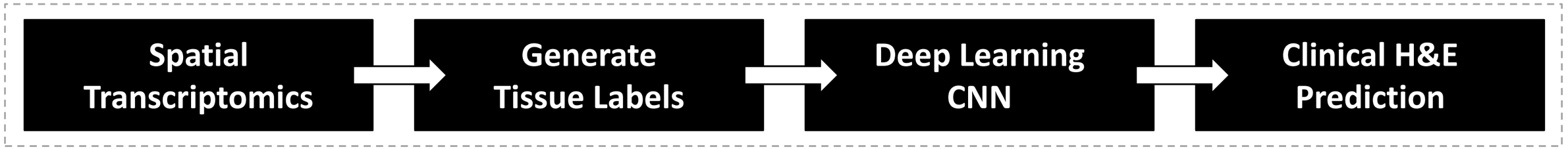
Spatial Transcriptomics (ST) yields unprecedented molecular architectures inside tissue.

- Cost-prohibitive in clinical practice.
- Requires highly specialized capture slides.
- Absent in routine pathology workflows.

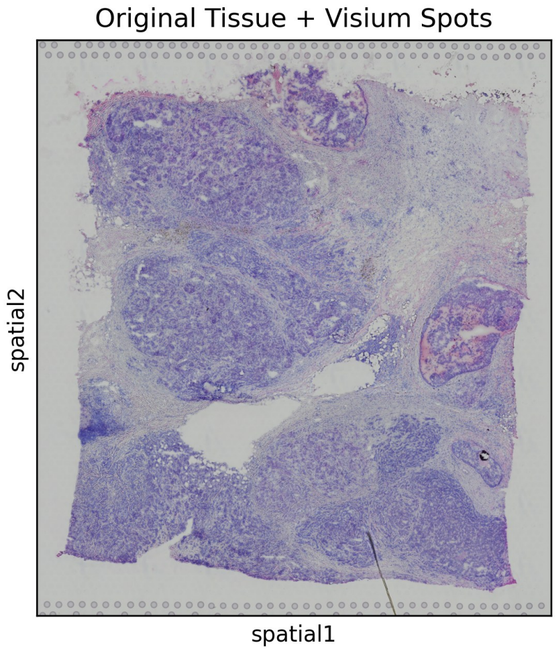
Our Approach

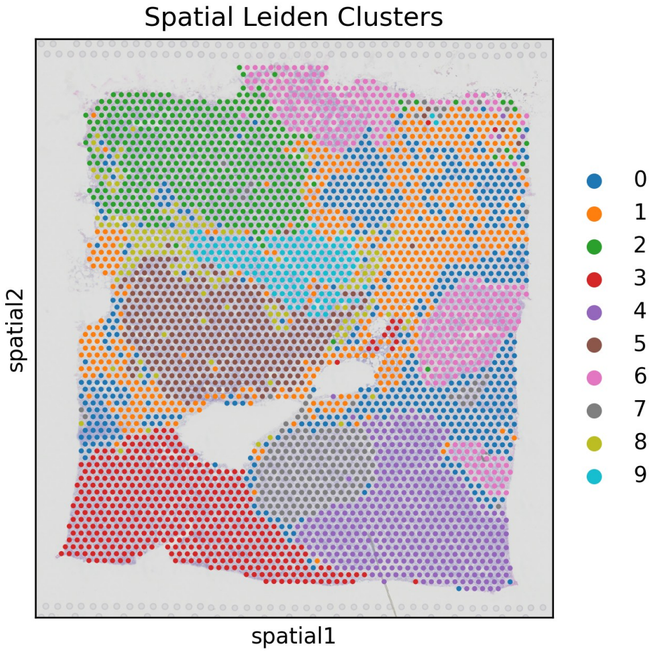
Convert high-dimensional ST assay into biological **training targets** for **deep learning** models.

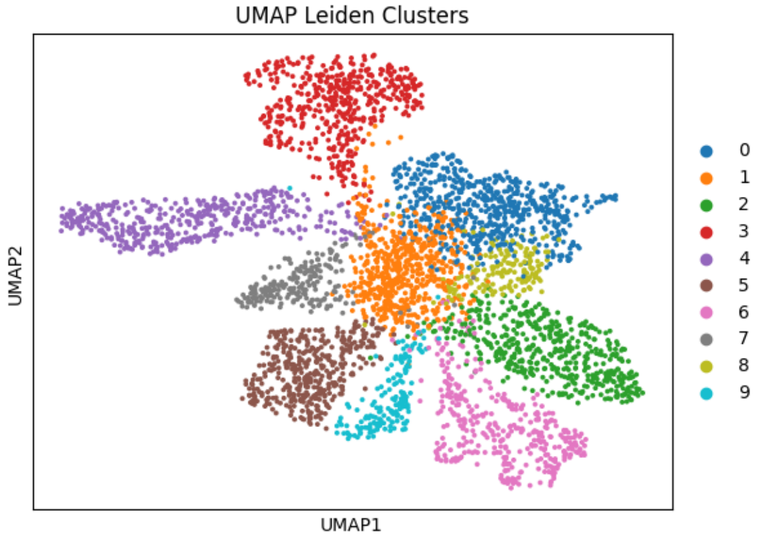
- Supervise models using real gene arrays.
- Predict directly from conventional H&E.
- Unlock complex profiling at zero extra cost.

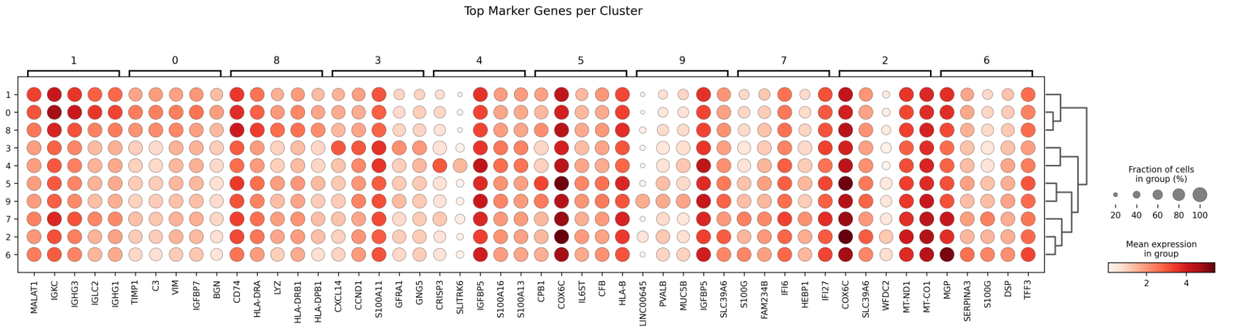


Using Spatial Transcriptomics as Ground Truth





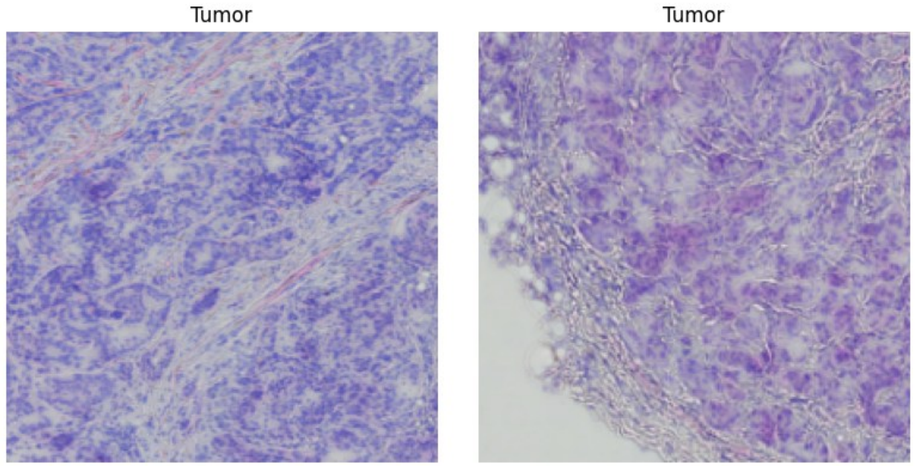


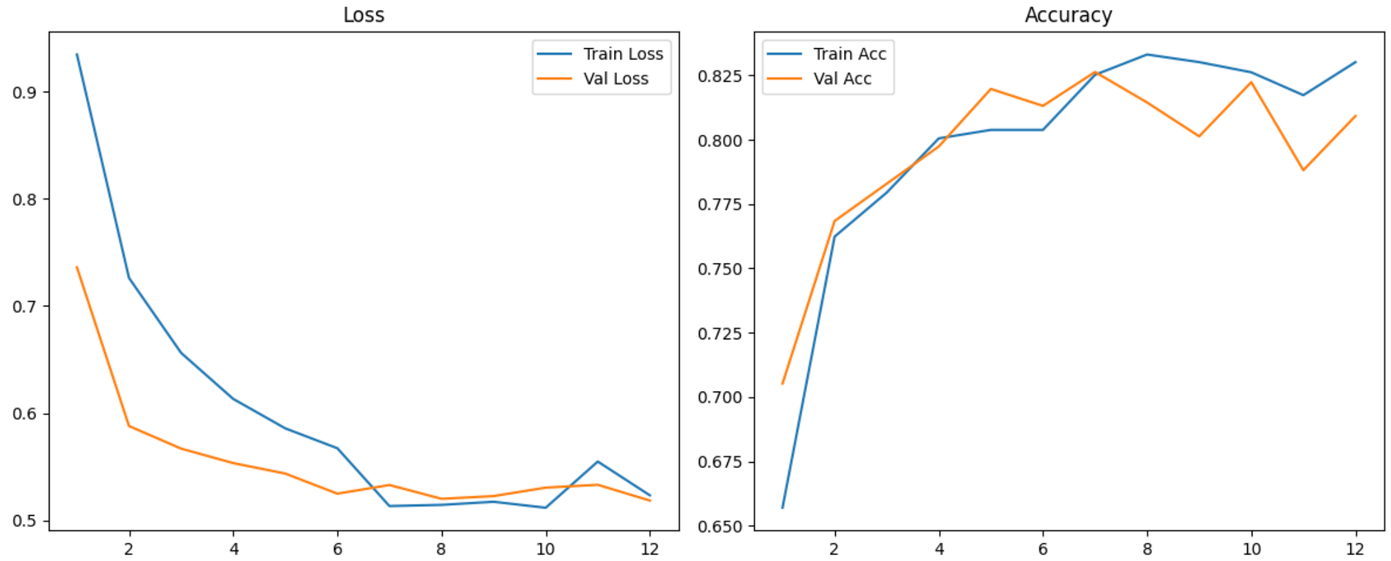


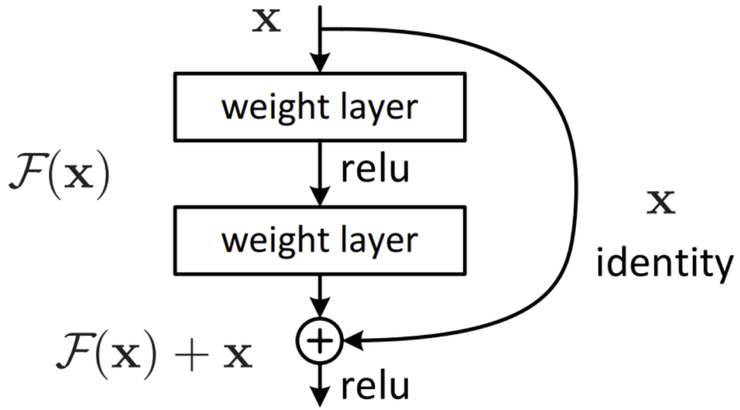
Leiden to Biological Domain Translation

We clusters spot gene-profiles unsupervised. Differentially expressed marker genes map each transcript cluster to concrete tumor, stroma, and immune zones.

Training a Histology-Based Classifier



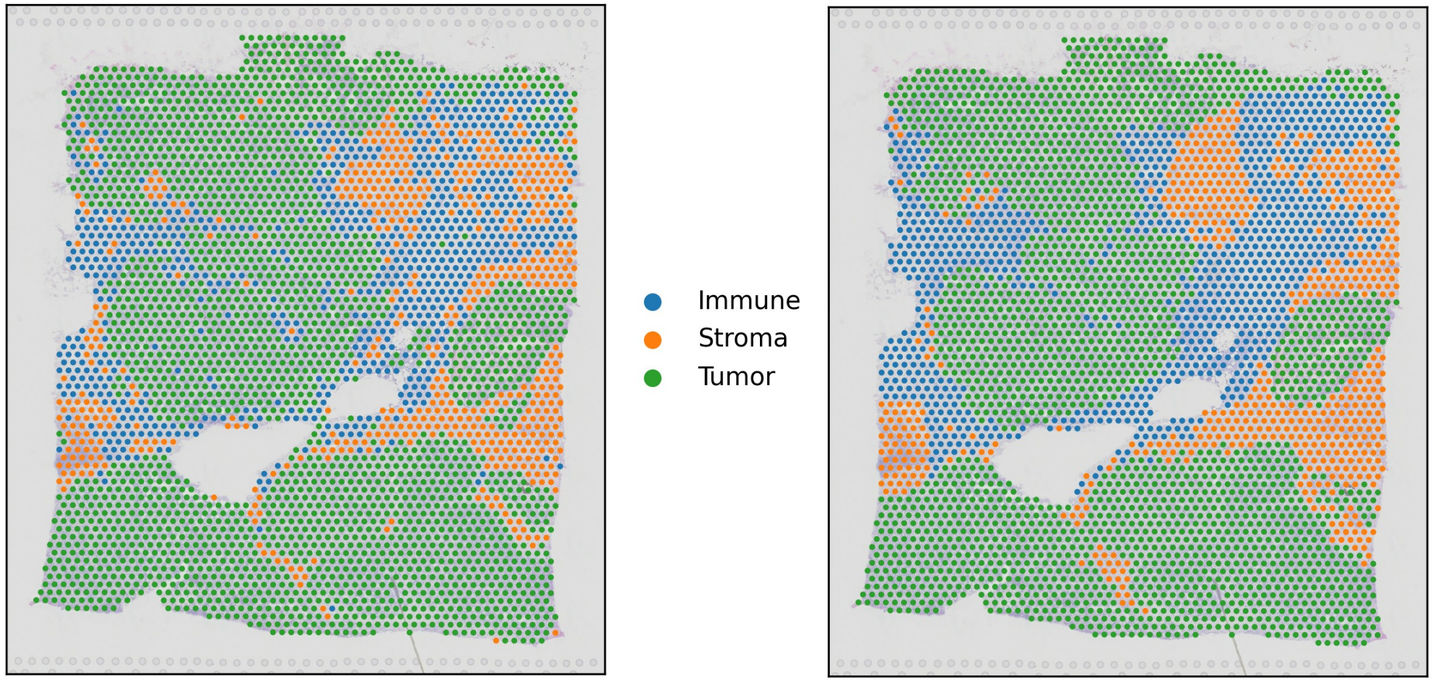




Model Convergence Overview

A pretrained ResNet18 backbone maps each 224×224 patch into dense feature representations, optimized using cross-entropy against genetic labels.

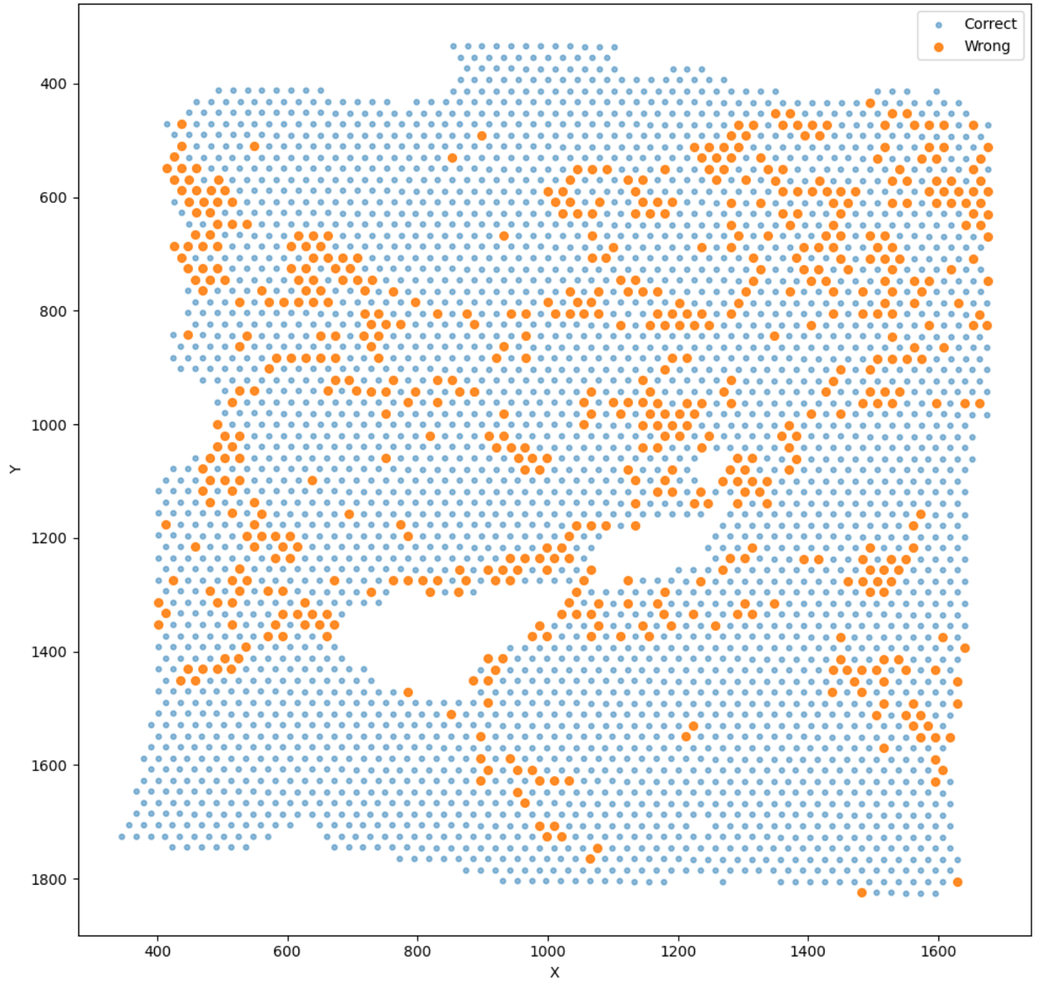
Spatial Prediction Results



Ground Truth Mask

Predicted TME Mask

Spatial Prediction Results



Core Domain Fidelity

- Accurately predicts tumor, stroma, and immune zones (blue).
- Proves the encoder successfully identifies tissue morphology.

Interface Boundary Gradients

- Disagreements (orange) concentrate at compartment margins.
- Reflects natural cellular gradients rather than rigid borders.

Key Takeaway

01. Molecular Ground Truths

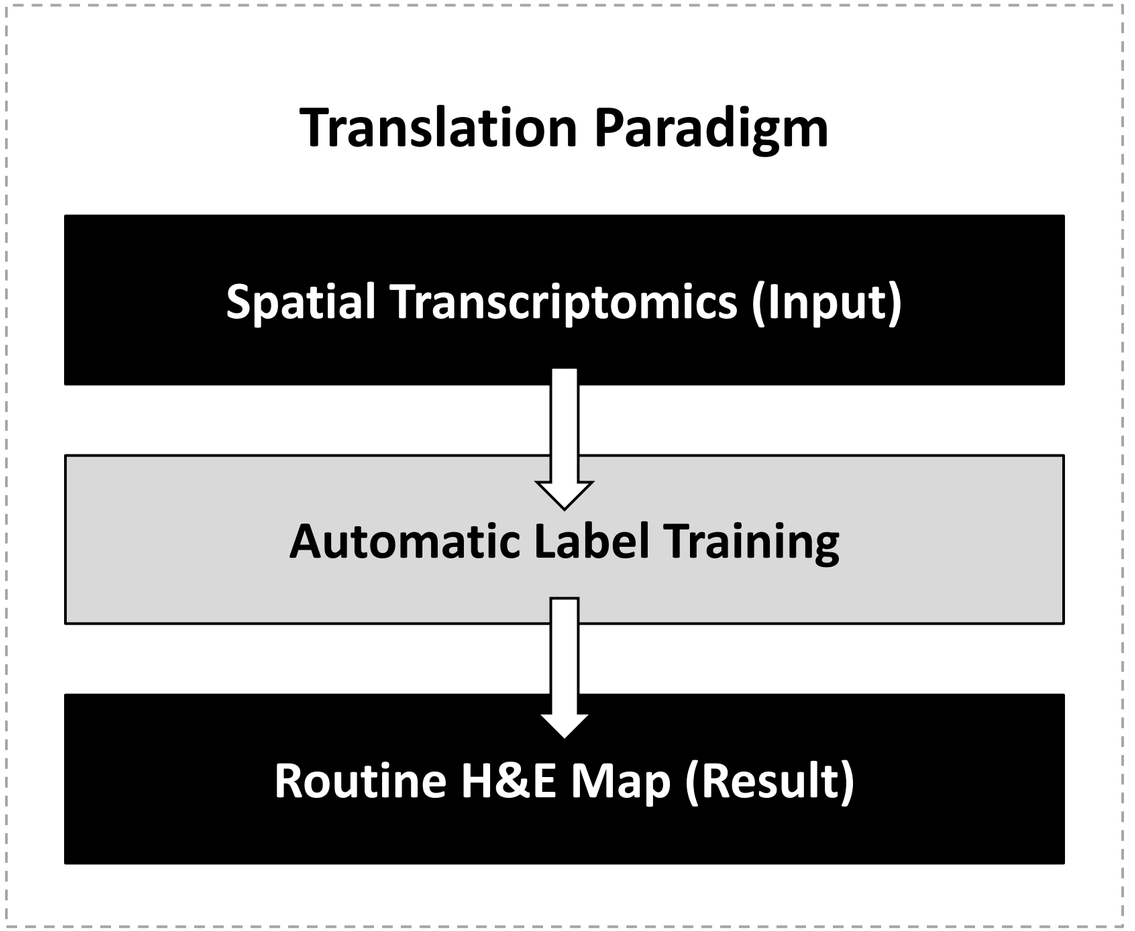
Bypasses manual pathology biases with precise, unsupervised spatial labels.

02. Structural Genetic Signals

Decodes complex cellular states from standard tissue features using simple AI.

03. Costless Diagnostics

Turns expensive sequencing data into routine, affordable slide diagnostics.



Future Directions

Data Augmentation	Improve model performance through advanced data augmentation strategies.
Clinical Validation	Perform clinical validation using larger and more diverse patient cohorts.
TME Subclustering	Investigate tumor microenvironment heterogeneity through spatial subclustering.
Immune Cell Subclustering	Characterize immune cell subpopulations and their spatial organization.
Quality Improvement	Extend the framework toward higher-resolution and single-cell-level spatial analysis.

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Thank You

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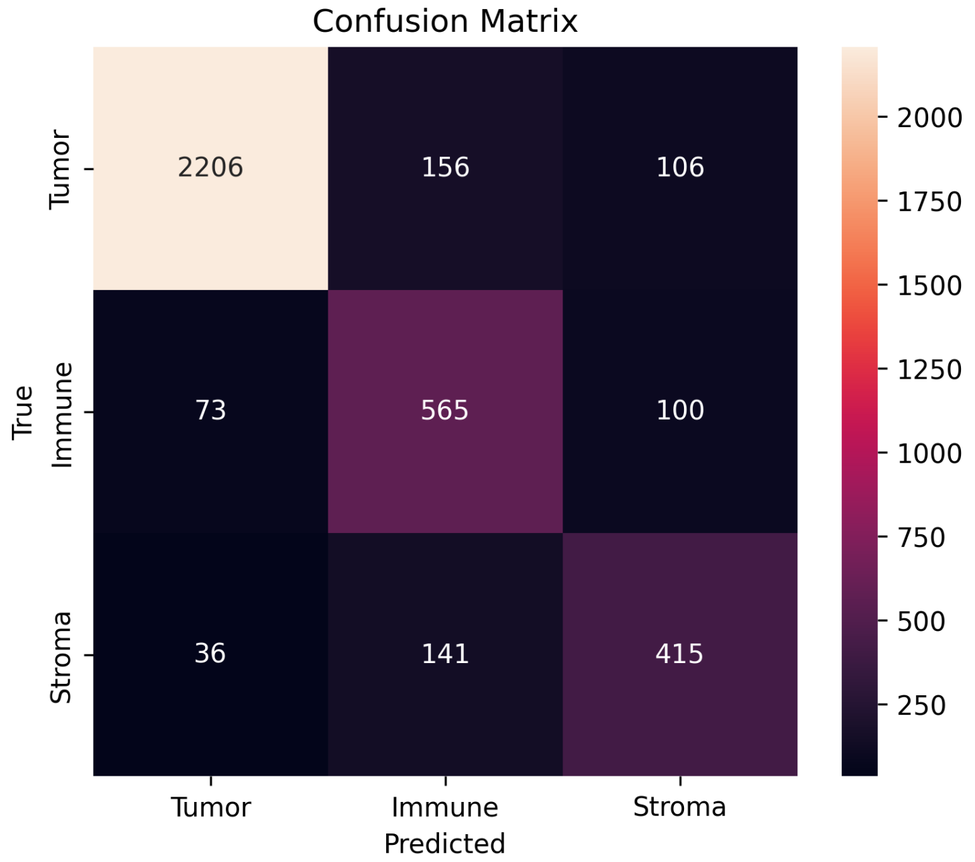
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GitHub Repository

Model Performance Summary

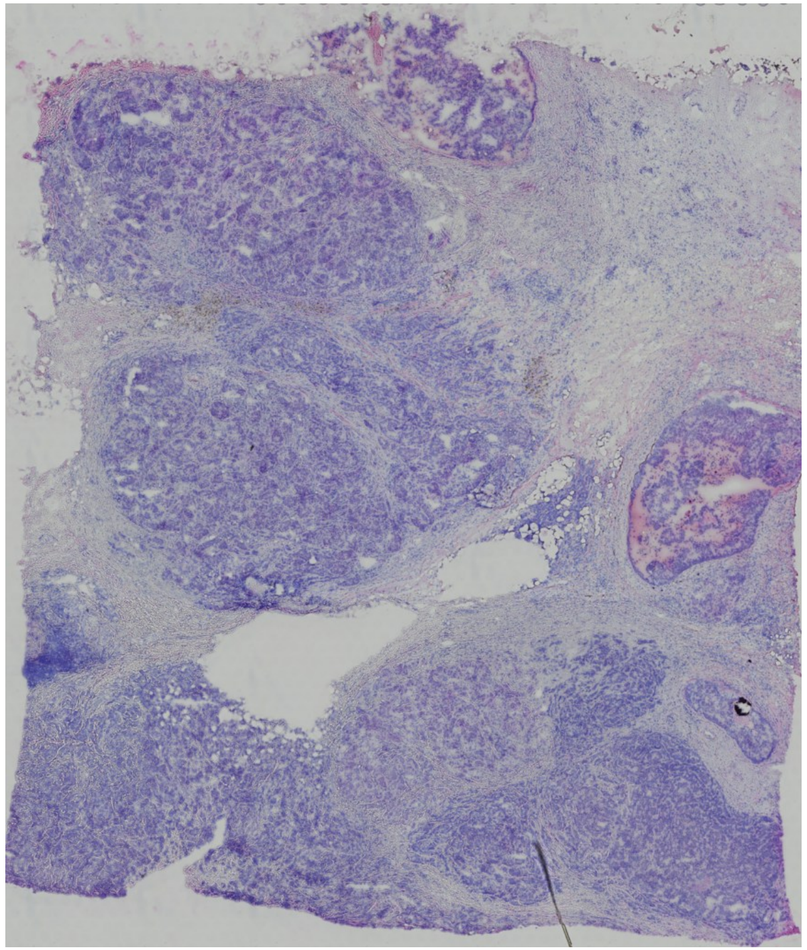


Class	Correct	Total	Accuracy
Tumor	2206	2468	89.4 %
Immune	565	738	76.6 %
Stroma	415	592	70.1 %
Overall	**3186**	**3798**	**83.9 %**

Key Observations

- 83.9% overall accuracy
- Best performance: Tumor (89.4%)
- Main confusion: Immune ↔ Stroma
- Supports histology-based TME prediction

Dataset Overview

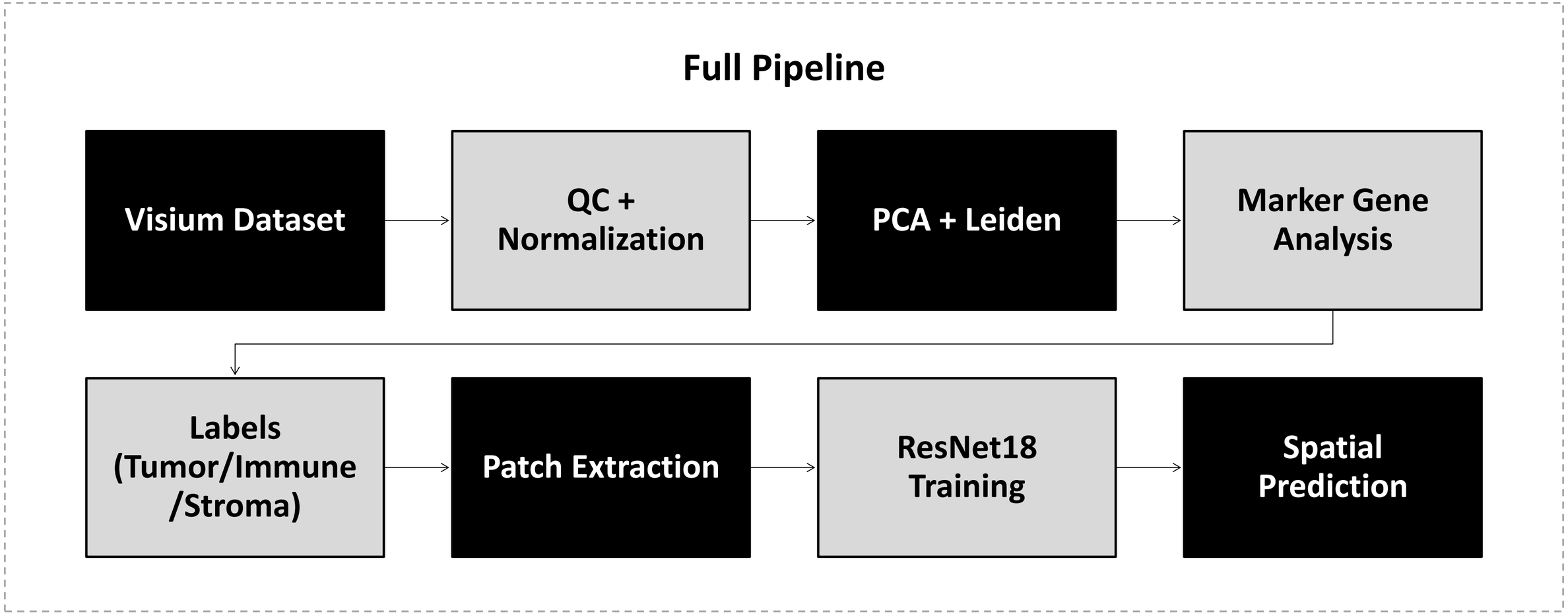


Item	Value
Dataset	10X Genomics Visium
Tissue	Breast Cancer
Number of Spots	3798
Number of Genes	21
Number of Patients	1

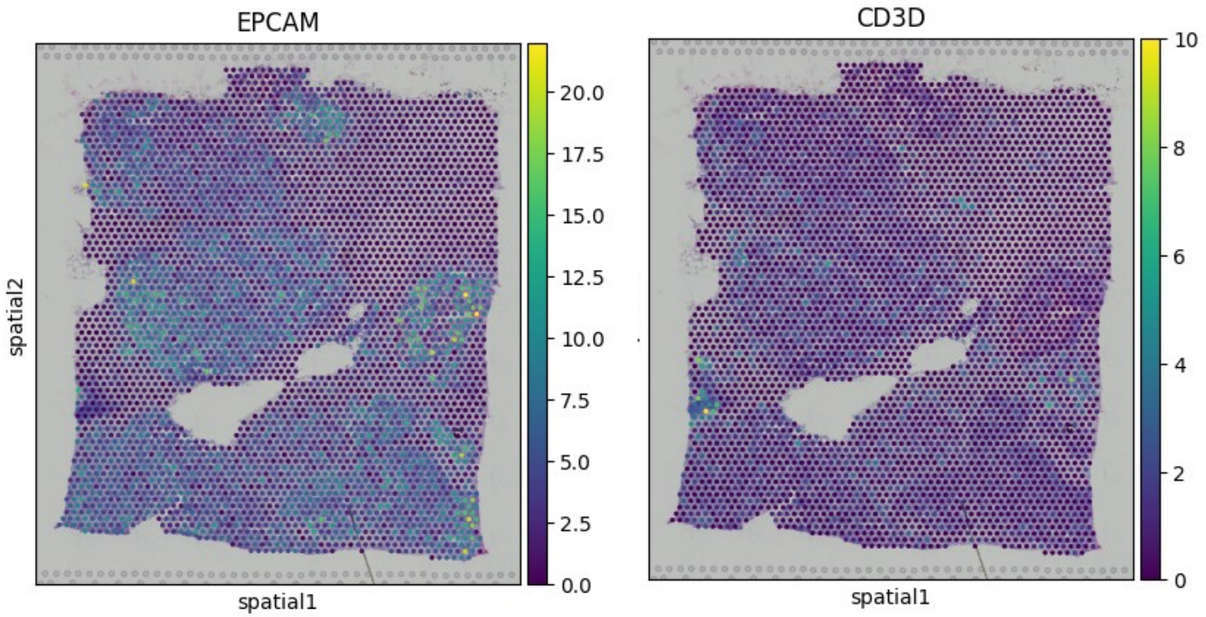
Dataset Summary

- Public 10x Genomics Visium breast cancer dataset
- 3,798 spatial spots with paired H&E image and gene expression
- Each spot converted into a 224 × 224 histology patch
- Single tissue section used as a proof-of-concept study

Complete Workflow



Ground Truth Construction



Tissue Type	Markers
Tumor	EPCAM, KRT8, KRT18, KRT19, EGFR, CDK4
Immune	CD3D, CD3E, CD3G, CD8A, CD8B, NKG7, GNLY, CD79A, MS4A1
Stroma	COL1A1, COL1A2, DCN, FAP, ACTA2, PDGFRB

Leiden Cluster → **Differential Expression** → **Marker Genes** → **Biological Mapping**

Current Limitations

Technical Limitations

- Single **Visium breast cancer sample** used for training and evaluation
- Train/validation split performed within the **same tissue section**
- No independent **external dataset validation** or cross-patient testing
- Relatively simple **ResNet18 architecture** compared with recent foundation models
- Limited assessment of **generalization to unseen histology images**

Biological Limitations

- Predicts only broad **Tumor, Immune, and Stroma** compartments
- Labels derived from **spot-level transcriptomic** measurements
- Unable to resolve **single-cell heterogeneity** within each spot
- Manual cluster-to-label mapping may introduce **biological interpretation bias**
- Does not capture finer TME populations such as **T-cell, B-cell, macrophage, or CAF subtypes**

Scope of This Study

Gene Prediction	Our Approach
Thousands of outputs	3 classes
Complex regression	Classification
Large datasets required	Suitable for proof-of-concept

Generalization to Unseen Data

Current Study

✓ Train/Validation Split

✗ Independent Patient Validation

✗ External Dataset Validation

Future Work

- Multiple Visium samples
- Cross-patient testing
- External cohorts